



I SEMESTER M.C.A

END SEMESTER EXAMINATIONS, DECEMBER 2023

SUBJECT: OBJECT ORIENTED PROGRAMMING [MCA 4121]

REVISED CREDIT SYSTEM

(02/12/2023)

Time: 3 Hours

MAX. MARKS: 50

Instructions to Candidates:

- ❖ Answer **ALL** the questions.
- ❖ Missing data may be suitably assumed.

1A.	Given a two-dimensional array of double type containing sales particulars of a shop as follows:	5																																			
	<table><tr><th>Day</th><th>Product1</th><th>Product2</th><th>Product3</th><th>Product4</th></tr><tr><td>MON</td><td>1750.75</td><td>233.90</td><td>540.00</td><td>567.75</td></tr><tr><td>TUE</td><td>1675.00</td><td>135.75</td><td>533.00</td><td>134.50</td></tr><tr><td>WED</td><td>233.90</td><td>1675.75</td><td>233.90</td><td>233.90</td></tr><tr><td>THU</td><td>1750.75</td><td>233.90</td><td>540.00</td><td>567.75</td></tr><tr><td>FRI</td><td>1675.75</td><td>233.90</td><td>233.90</td><td>1675.00</td></tr><tr><td>SAT</td><td>135.75</td><td>533.00</td><td>134.50</td><td>233.90</td></tr></table>	Day	Product1	Product2	Product3	Product4	MON	1750.75	233.90	540.00	567.75	TUE	1675.00	135.75	533.00	134.50	WED	233.90	1675.75	233.90	233.90	THU	1750.75	233.90	540.00	567.75	FRI	1675.75	233.90	233.90	1675.00	SAT	135.75	533.00	134.50	233.90	
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	Write a program to perform the following operations: i) Find the total daily sales on each day. ii) Find the weekly sale of each product iii) Find the average weekly sales of shop.																																				
1B.	Consider a class named Book_Info that stores information about a book, including the book ID (integer), book title (character array), number of copies (integer), and price (float). Define <i>default</i> , <i>parameterized</i> and <i>copy constructors</i> for initializing objects of this class.	3																																			
1C.	Write C++ statements to dynamically allocate memory for a 2-dimensional array of type double with 3 rows and 4 columns.	2																																			
2A.	What are the steps to implement <i>pure virtual function</i> and achieve runtime polymorphism? Provide a real-world example to illustrate these concepts.	5																																			
2B.	Write a C++ program that includes a class called My_Time with data members for hours, minutes, and seconds. Define a friend function to overload the unary increment operator 'T++' for an object 'T' of the My_Time class, which increments the time by 1 hour.	3																																			
2C.	Mention any two uses of ' <i>this</i> ' pointer.	2																																			
3A.	Generally, the size of a file may be measured in either bytes or kilobytes. Design a data conversion program to convert input file size from kilobytes to bytes and vice versa. In this program, kilobytes	5																																			

	should be considered as the user-defined data type, and bytes should be represented by the basic type 'long'.	
3B.	Discuss the steps involved in handling <i>user-defined exception</i> .	3
3C.	Differentiate between <i>structure</i> and <i>union</i> .	2
4A.	Consider a class called <i>Patient</i> which stores the information about each patient such as: patient id (integer, starting from 1), patient name (string), age (integer), gender(character). Create the above class with necessary input & output functions and store the objects of this class in a file called 'Patient_info.dat'. You need to display the details of a particular patient based on the patient id entered by the user using the concept of <i>random-access file</i> . Write a complete program to implement the solution to the above problem.	5
4B.	Write a <i>template function</i> that displays the total number of positive and negative numbers in each column of a two-dimensional array. The function should take the array name and its dimensions (of type integer) as arguments. In the main () function, demonstrate the usage of the template function with arrays of type int, float, and double.	3
4C.	Given two objects of <i>ofstream</i> and <i>ifstream</i> classes, namely, op_file and ip_file respectively. Write statements to perform the following operations: i) To move the get pointer by 7 bytes in the backward direction from the current position. ii) To move the put pointer to the end of the file.	2
5A.	How do you display the contents of a vector in both forward and backward directions using <i>iterators</i> and <i>reverse iterators</i> ? Explain with the help of a suitable example.	5
5B.	Create a function <i>check_email(char email[])</i> to check whether the email address ends with manipal.edu or not. In the main() function, declare a function pointer and invoke the <i>check_email</i> function using this function pointer to determine if the given email address meets the specified criteria.	3
5C.	Briefly explain (with syntax and example) any two member functions of <i>string</i> class.	2
